$$\Gamma = \frac{(R_S+2Z_C)-2Z_C}{(R_S+2Z_C)+2Z_C} = \frac{R_S}{R_S+2Z_C}$$

$$\Rightarrow \left|\frac{V_{IM}}{V_{OUT}}\right| = \left|\frac{R_S+2Z_C}{R_S}\right| = 1+\frac{2Z_C}{R_S}$$

$$A = 20\lg\left(1+\frac{2Z_C}{R_S}\right)$$

Varianta pt atenuări mici

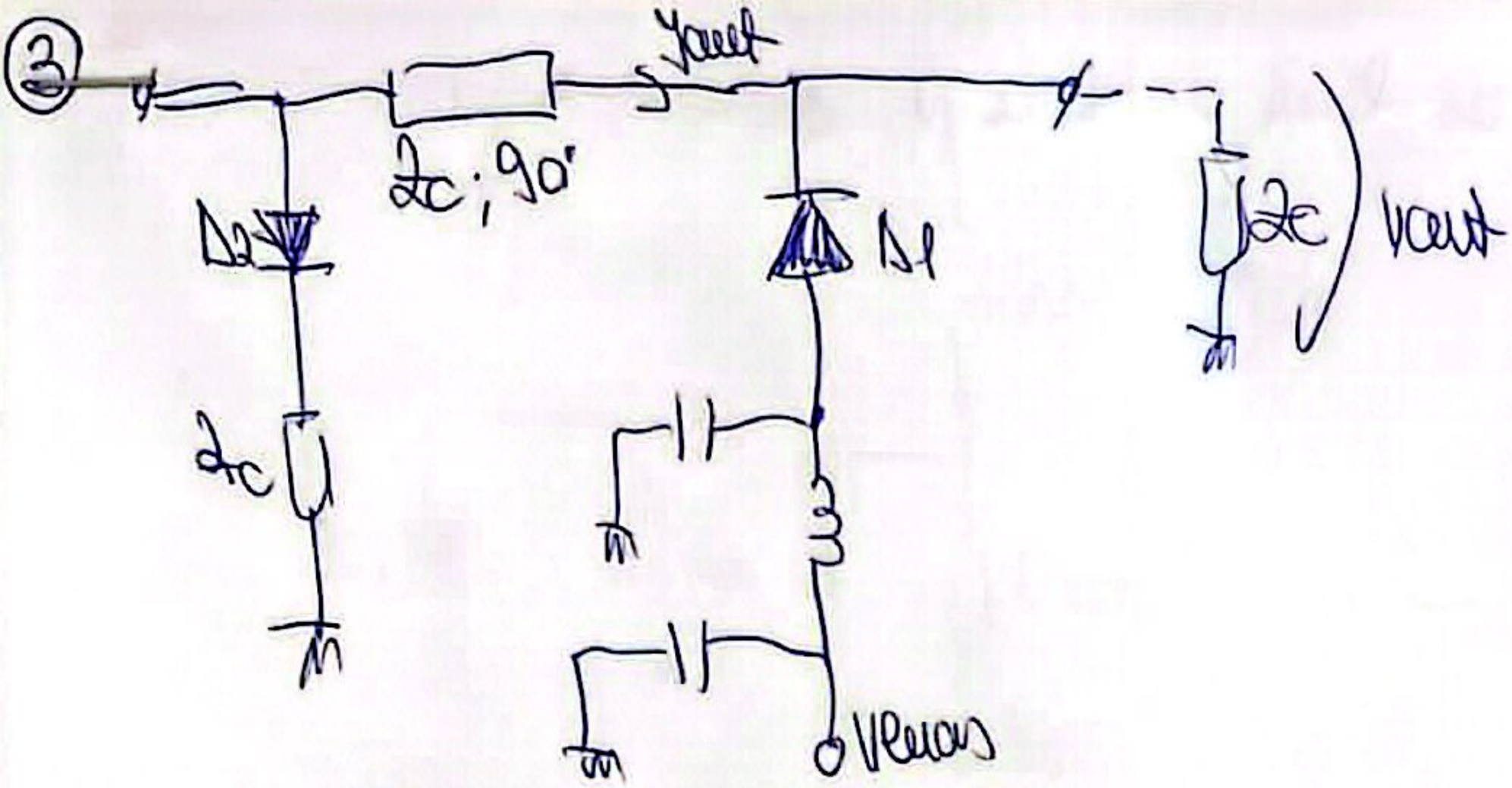
$$V_{r,IM} = \frac{V_{IM}}{\sqrt{2}} e^{j0} \frac{\Gamma}{\sqrt{2}} + \frac{V_{IM}}{\sqrt{2}} e^{-j90^\circ} \frac{\Gamma}{\sqrt{2}} e^{-j90^\circ}$$

↳ tens refl la intrare

$$= \frac{V_{IM}\Gamma}{2} + \frac{V_{IM}\Gamma}{2} \underbrace{e^{-j180^\circ}}_{-1} = 0.$$

- Atenuator cu diferă P_{IM} pt at mărit -

③



$D1 = D2$ (în conducție directă)

$V_{bias} > 0$.

$$V_{in} = \underbrace{V_{out}\cos\theta}_{0} + \underbrace{j Z_c \sin\theta}_{1}\, \underbrace{Y_{out}}_{\frac{V_{out}}{Z_c \| R_S}}$$

$$V_{in} = j\cancel{Z_c}\,\frac{V_{out}}{\frac{\cancel{Z_c} R_S}{Z_c + R_S}} = j V_{out}\,\frac{Z_c + R_S}{R_S}$$

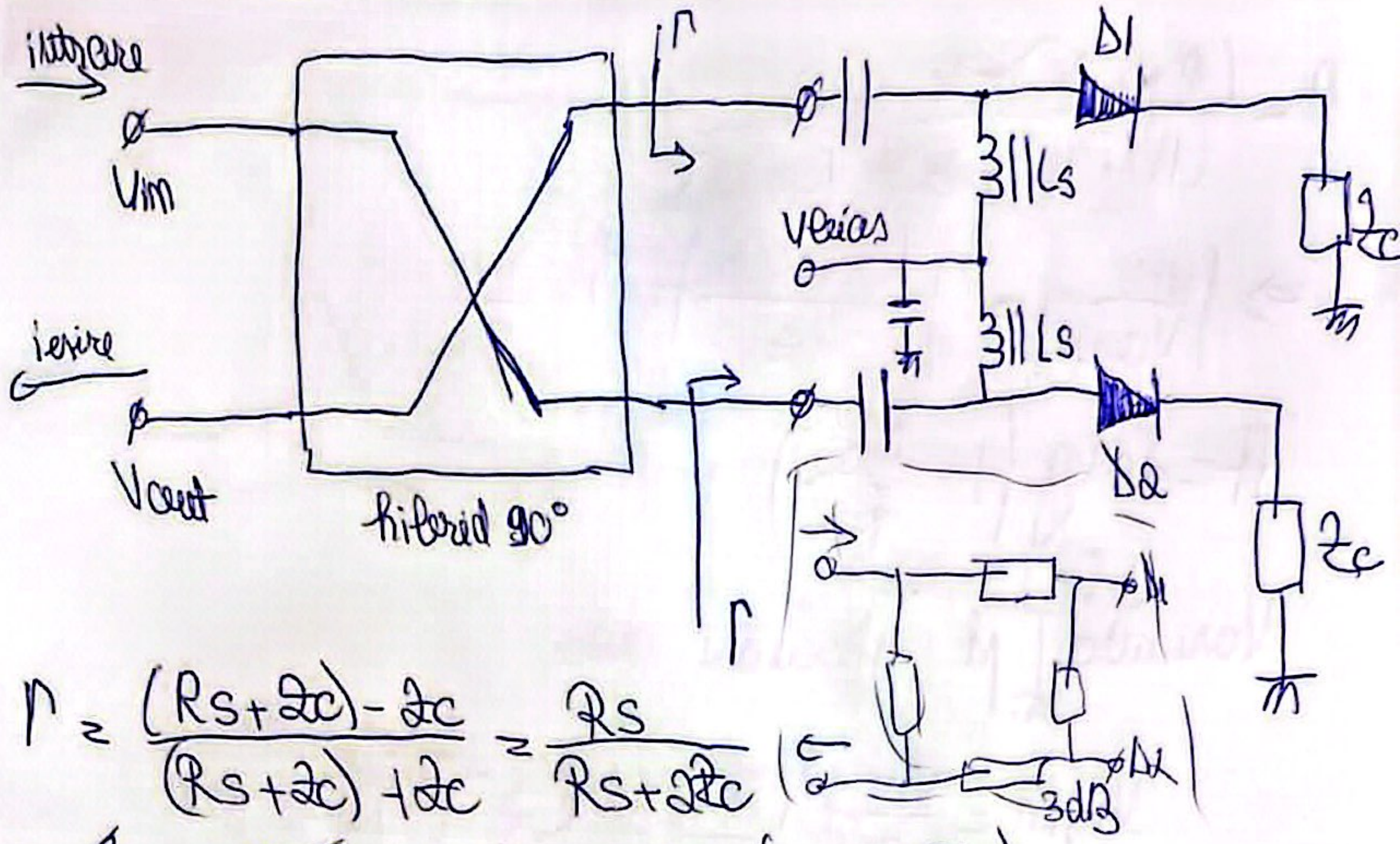
$$A(dB) = 20\lg\left|\frac{V_{in}}{V_{out}}\right| = 20\lg\left(1 + \frac{Z_c}{R_S}\right) \Rightarrow$$ pt cât mari

$$Z_{in} = (Z_c + R_S) \| \left(\frac{Z_c^2}{R_S \| Z_c}\right) =$$

adaptare pt lumea de imped ? la intrare

$$= \frac{\cancel{(Z_c + R_S)}\,\frac{Z_c^2(R_S + Z_c)}{R_S \cdot Z_c}}{\cancel{(Z_c + R_S)} + \frac{Z_c^2\cancel{(R_S + Z_c)}}{R_S \cdot Z_c}} = \frac{Z_c^2(Z_c + R_S)}{Z_c^2 + R_S \cdot Z_c} = Z_c$$

1.



$$\Gamma = \frac{(R_s+Z_c)-Z_c}{(R_s+Z_c)+Z_c} = \frac{R_s}{R_s+2Z_c}$$

$V_{polas} \rightarrow R_s \nearrow$; D1 și D2 identice (diode PIN)

R_s - rezistența serie a diodelor

$V_{polas} > 0$

Dacă $R_s \cong 0$ ($R_s \ll Z_c$) $\rightarrow V_{out} = 0 \Rightarrow A \rightarrow \infty$

$$A[dB] = -10\lg\left(\frac{P_{out}}{P_{in}}\right) = -20\lg\left|\frac{V_{out}}{V_{in}}\right| =$$

$$= 20\lg\left|\frac{V_{in}}{V_{out}}\right|$$

Dacă $R_s \neq 0 \Rightarrow V_{out} \neq 0 \Rightarrow A =$ finită

$$V_{out} = \frac{V_{in}}{\sqrt{2}}\Gamma \cdot \frac{1}{\sqrt{2}} e^{-j90^\circ} + \frac{V_{in}}{\sqrt{2}} e^{-j90^\circ} \cdot \Gamma \cdot \frac{1}{\sqrt{2}} =$$

$$= 2 \cdot \frac{\Gamma}{2} \cdot V_{in} e^{-j90^\circ} = \Gamma e^{-j90^\circ} \cdot V_{in}$$

$$\left|\frac{V_{in}}{V_{out}}\right| = \frac{1}{|\Gamma|}$$

- Atenuator cu linii de transm pt atenuări mari -

Scanned with CamScanner

- Var pt alternare mici cu diode pin -

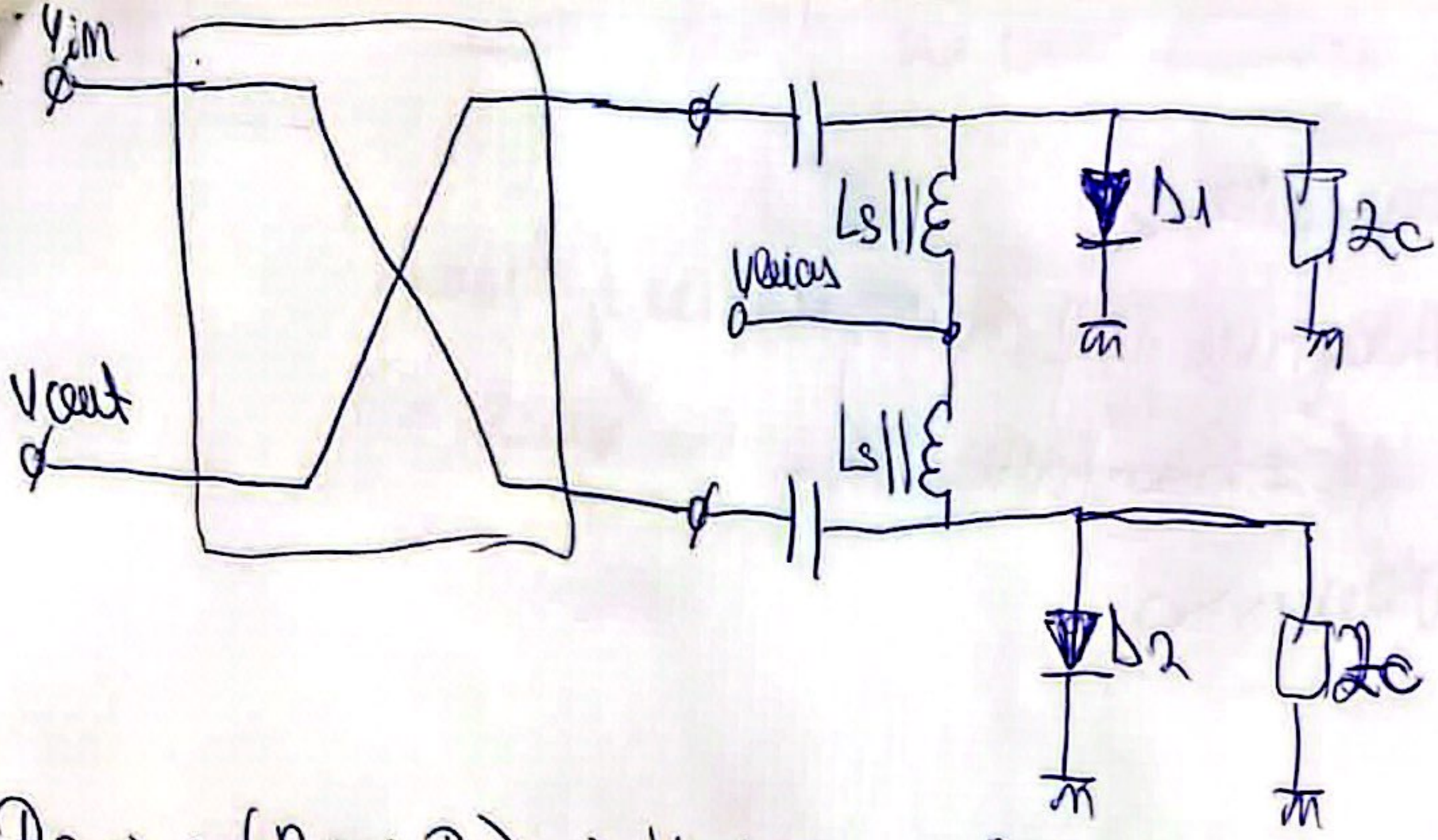
Observatii:

Adaptare perf de impedanta ($V_{rim}=0$)

R_S = Rezistenta serie a diodelor

Uneori >0.

③



$R_S \to \infty \; (R_S \,||\, Z_C) \to V_{out} = 0 \Rightarrow A \to \infty$

Dacă $R_S \neq \infty \Rightarrow V_{out} \neq 0 \Rightarrow A =$ finită

$$\left|\frac{V_{im}}{V_{out}}\right| = \frac{1}{|\Gamma|} = \mu$$

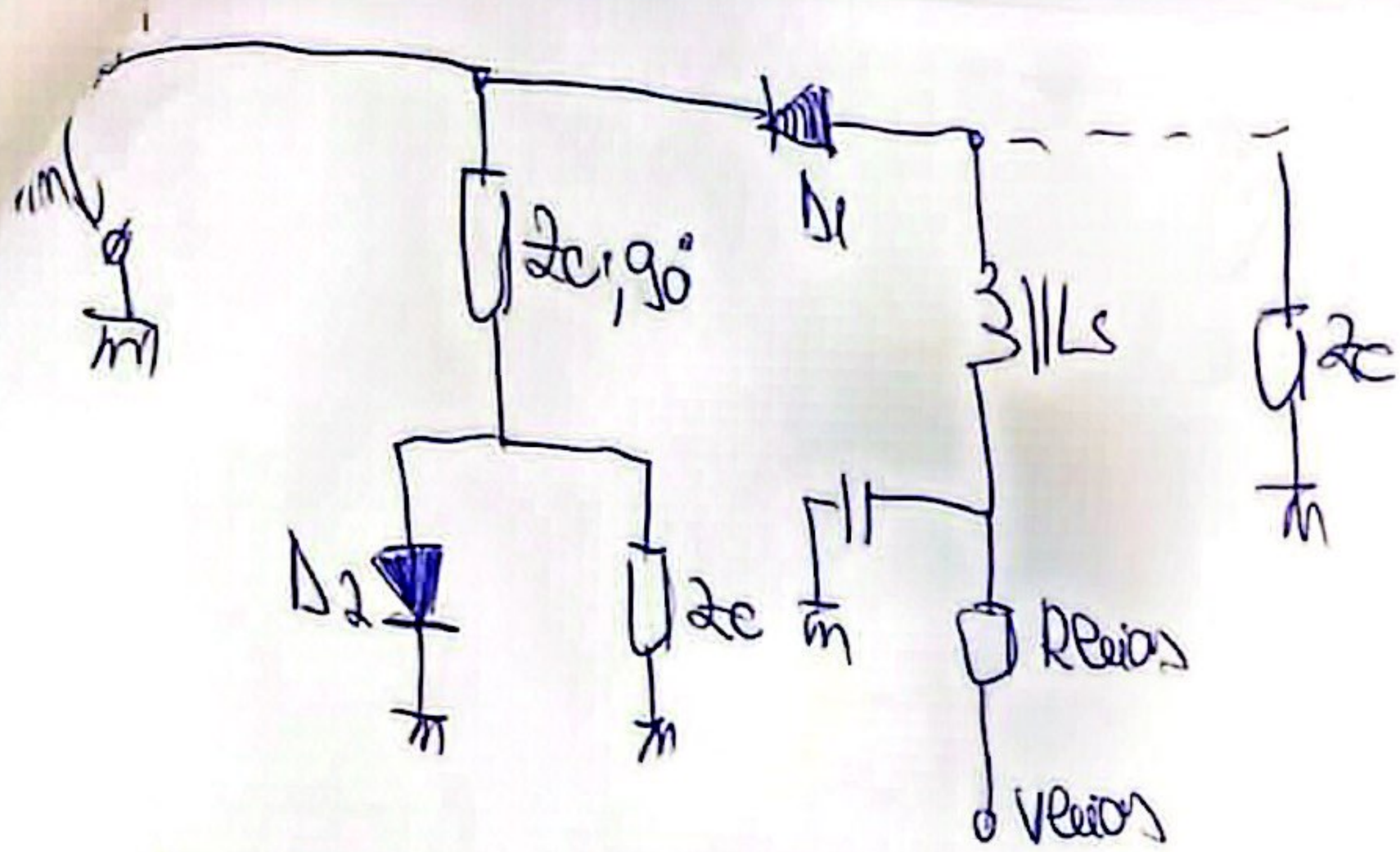
$$\boxed{\begin{aligned}\Gamma &= \frac{(R_S \,||\, Z_C) - Z_C}{(R_S \,||\, Z_C) + Z_C} = \frac{\frac{R_S Z_C}{R_S + Z_C} - Z_C}{\frac{R_S Z_C}{R_S + Z_C} + Z_C} = -\frac{Z_C^2}{2R_S Z_C + Z_C^2} \\ &= -\frac{Z_C}{2R_S + Z_C}\end{aligned}}$$

$$\mu = \left|-\frac{2R_S + Z_C}{Z_C}\right| = 1 + \frac{2R_S}{Z_C}.$$

$$\Rightarrow A = 20 \log\left(1 + \frac{2R_S}{Z_C}\right)$$

$$V_{r,im} = \frac{V_{im}}{\sqrt{2}} e^{j0} \Gamma \cdot \frac{1}{\sqrt{2}} + \frac{V_{im}}{\sqrt{2}} e^{-j90^\circ} \frac{\Gamma}{\sqrt{2}} e^{-j90^\circ} =$$

$$= \frac{V_{im}\Gamma}{2} + \frac{V_{im}\Gamma}{2} \cdot e^{-j180^\circ} = 0$$



D1 = D2

D1/2 în conducție directă (Vsarcin > 0)

$$Z_{in} = (R_S + Z_C) \parallel \left(\frac{Z_C^2}{R_S \parallel Z_C}\right) =$$

adaptare fct bună de imped la intrare

$$= \frac{(R_S + \cancel{Z_C})\,\dfrac{Z_C^2(R_S+Z_C)}{R_S \cdot Z_C}}{(R_S + \cancel{Z_C}) + \dfrac{Z_C^2(R_S+\cancel{Z_C})}{R_S \cdot Z_C}} = \frac{Z_C^2(R_S+Z_C)}{R_S Z_C + Z_C^2} = Z_C$$

$$V_{out} = \frac{Z_C}{Z_C + R_S} \cdot V_{in} \Rightarrow A(dB) = 20 \lg\left(1 + \frac{R_S}{Z_C}\right) \rightarrow \mu t$$

atenuări mici